

Year 8 Test

Term 3

Equations, Statistics and Time



Baldivis
Secondary College

Resources Allowed

- 1 A4 page of notes
- Calculator

Name: Solns.

Score: / 54

1. [16 marks]

For each of the following equations find the value of x .

a) $x + 4 = 12$

$x = 8$ ✓

b) $2x = 24$

$x = 12$ ✓

c) $3x + 2 = 17$

$3x = 15$ ✓
 $x = 5$ ✓

d) $\frac{x}{3} - 7 = 2$

$\frac{x}{3} = 9$ ✓
 $x = 27$ ✓

e) $\frac{x-2}{2} + 2 = 35$

$\frac{x-2}{2} = 33$ ✓
 $x-2 = 66$ ✓
 $x = 68$ ✓

f) $3(x + 12) = 18$

$x + 12 = 6$ ✓
 $x = -6$ ✓

g) $3x = x + 4$

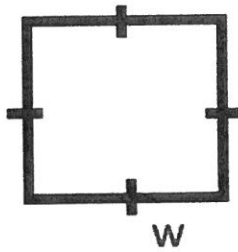
$2x = 4$ ✓
 $x = 2$ ✓

h) $x = \frac{2x-4}{4}$

$4x = 2x - 4$ ✓
 $2x = -4$ ✓
 $x = -2$ ✓

2. [3 marks]

A square has a perimeter of 36cm.



- a) Formulate an equation using w as the width,

$$4w = 36$$

✓

- b) Using your equation from a), determine the width of the square.

$$w = \underline{9 \text{ cm}}$$

✓✓

3. [3 marks]

A number is tripled, and then 2 is added. This gives the same result as if the number were quadrupled. Setup and solve an equation to find the original number.

$$\begin{aligned} 3x + 2 &= 4x \\ 2 &= x \end{aligned}$$

4. [4 marks]

Jeremy writes down how many hours he works each day for one week.

Day	Monday	Tuesday	Wednesday	Thursday	Friday
Number of hours	8	10	8	7	9

a) What is the mean number hours Jeremy worked each day?

$$\frac{42}{5} = 8.4$$

b) What is the median number of hours Jeremy worked each day?

8

c) What is the mode number of hours Jeremy worked each day?

8

5. [9 marks]

Consider the following set of data

73, 57, 73, 62, 20, 69, 62, 73, 65, 67

20, 57, 62, 62, 65, 67, 69, 73, 73, 73

a) Calculate;

I. Mean

$$\frac{548}{10} = 54.8$$

II. Median

66

III. Mode

73

IV. Range

$$73 - 20 = 53$$

b) Which score in the set of data is an outlier?

20

c) If the outlier is removed, what effect will this have on the;

I. Mean

Increase

II. Median

Increase to 67

III. Mode

No change

IV. Range

decrease to 16

6. [3 marks]

Fill in the blanks of the following sentences using words from this list.

Survey, Sample, Data, Bias, Census

A Sample is when part of a population is surveyed.

When the entire population is surveyed this is called a Census.

When a sample doesn't reflect the entire population it is said to have Bias.

7. [4 marks]

State whether the following are a Sample or a Census

a) A random selection of some people at a shopping centre

Sample.

b) A stocktake of the goods in a store

Census.

c) A crash test of new cars manufactured by a factory

Sample.

d) Asking all the Teachers at your school whether they approve of a new class timetable

Census.

8. [6 marks]

Complete the table

24 hour time	12 hour time
0017	12:17am
0915	9:15am
0835	8:35am
1440	2:40pm
1959	7:59pm
2130	9:30pm

9. [2 marks]

Jarod went to the movies with some friends. If the movie started at 7:00pm and the movie went for 1 hour and 30 mins, at time did the movie finish?

8:30pm.

✓✓

10. [2 marks]

A taxi departs at 18:52 from Sydney and arrives at 21:33 at Melbourne. How long did the taxi take for the trip?

2hr 41mins.

✓✓

11. [2 marks]

Sarah in Sydney is expecting a phone call from Fiona in Manilla. Manilla is 2 hours behind Sydney time.

a) If Sarah calls at 10:45am, what time is it for Fiona?

8:45am.

✓

b) If Fiona calls at 3:30pm, what time is it for Sarah?

5:30pm.

✓

END OF TEST

